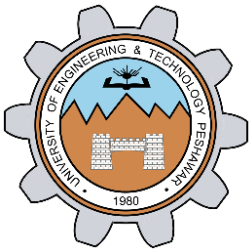
Interfacing Seven Segment Display to 8051 Development kit

LAB # 05



Spring 2024

CSE-307L Microprocessor Based System Design Lab

Submitted by: **Ali Asghar**

Registration No.: **21PWCSE2059**

Class Section: **C**

"On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work."

Submitted to:

Dr. Asif Ali Khan

Date:

29th March 2024

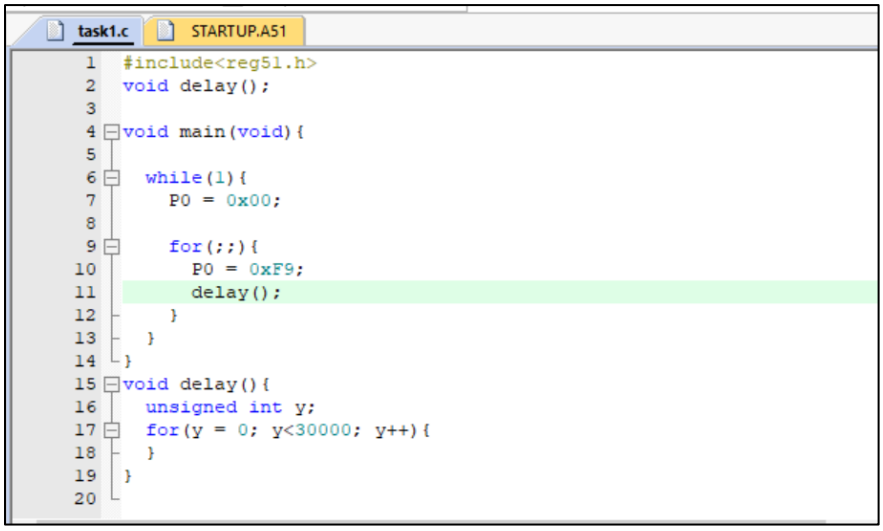
Department of Computer Systems Engineering

University of Engineering and Technology, Peshawar

Task 1:

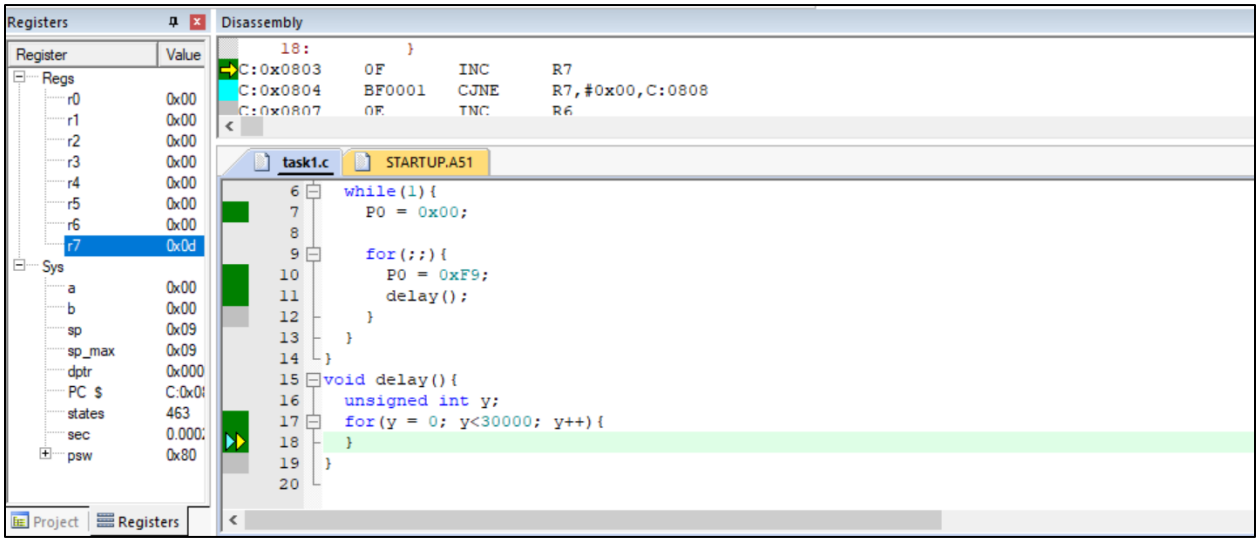
Run all the programs given in the lecture

Code:

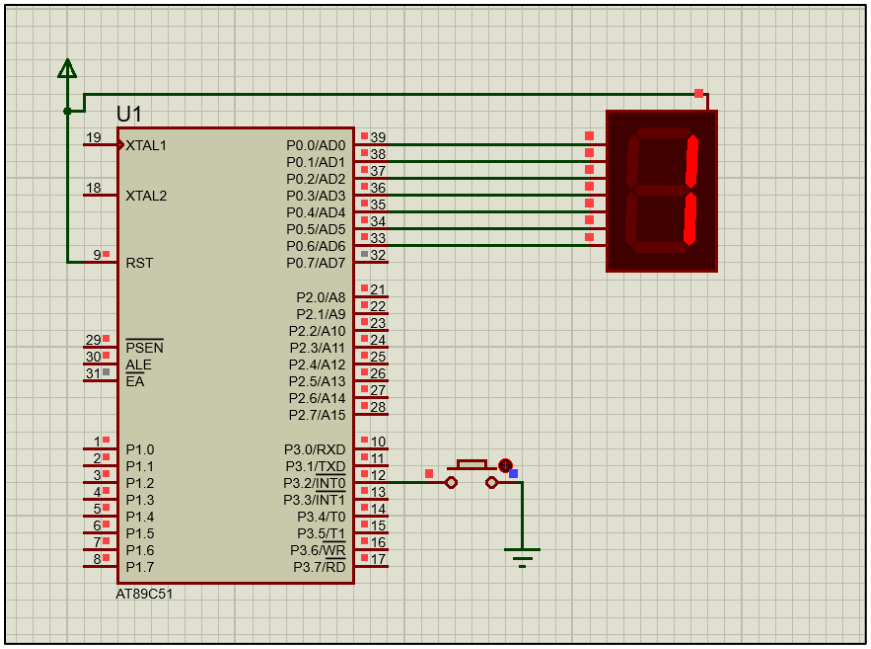


```
#include<reg51.h>
void delay();

void main(void){

  while(1){
    P0 = 0x00;

    for(;;){
      P0 = 0xF9;
      delay();
    }
  }
}
void delay(){
  unsigned int y;
  for(y = 0; y<30000; y++){
  }
}
```

Output:



Proteus Simulation:



Task 2:

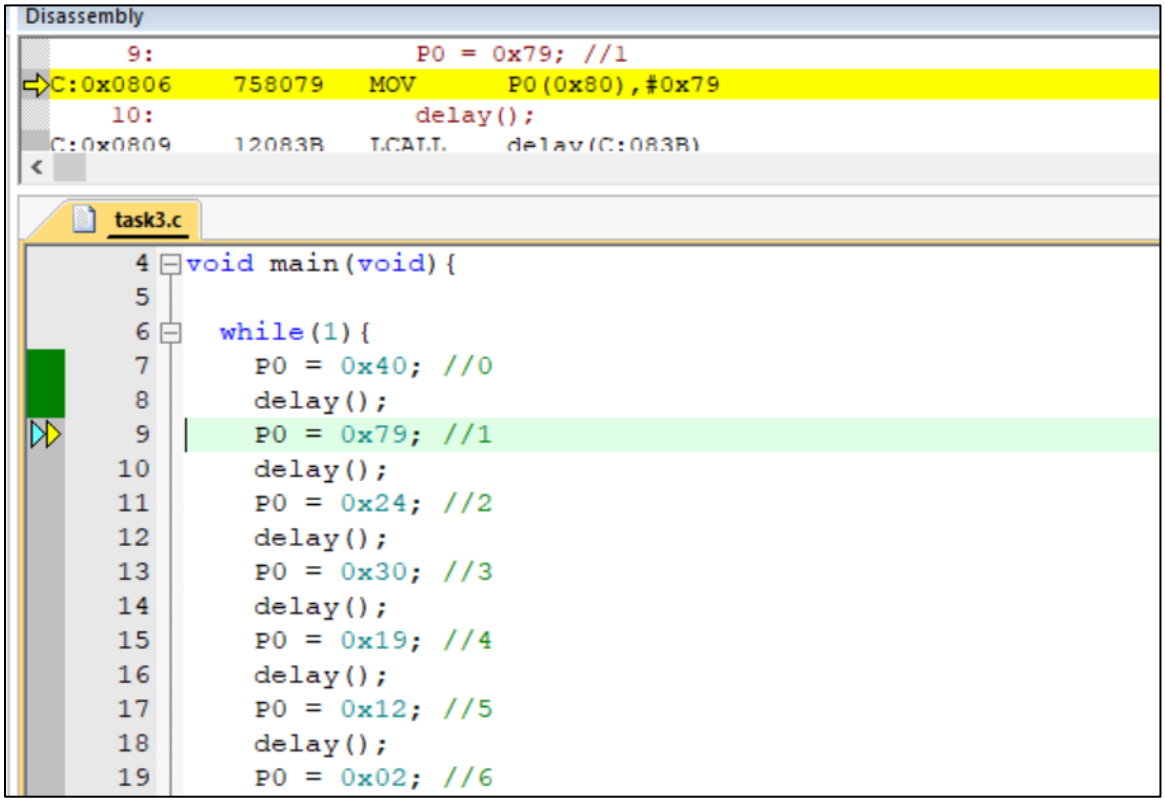
Display different digits on the Seven Segment Display

Code:

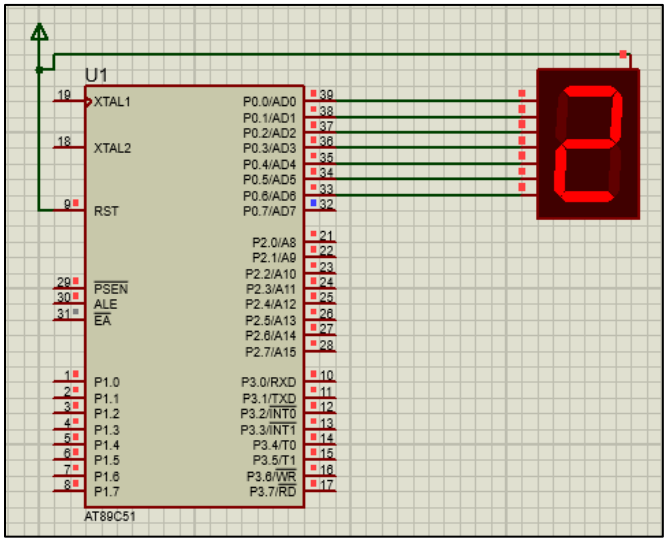
```
#include<reg51.h>
void delay();

void main(void){

  while(1){
     P0 = 0x40; //0
     delay();
     P0 = 0x79; //1
     delay();
     P0 = 0x24; //2
     delay();
     P0 = 0x30; //3
     delay();
     P0 = 0x19; //4
     delay();
     P0 = 0x12; //5
```

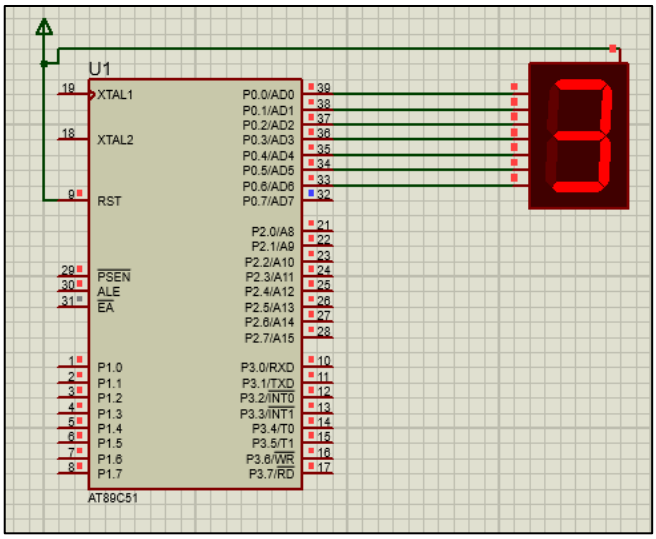
```
16      delay();
17      P0 = 0x12; //5
18      delay();
19      P0 = 0x02; //6
20      delay();
21      P0 = 0x78; //7
22      delay();
23      P0 = 0x00; //8
24      delay();
25      P0 = 0x18; //9
26    }
27  }
28  void delay(){
29    unsigned int y, x;
30    for(x = 0; x<30000; x++);
31    for(y = 0; y<30000; y++);
32  }
33
```

Output:



Proteus Simulation:





Task 3:

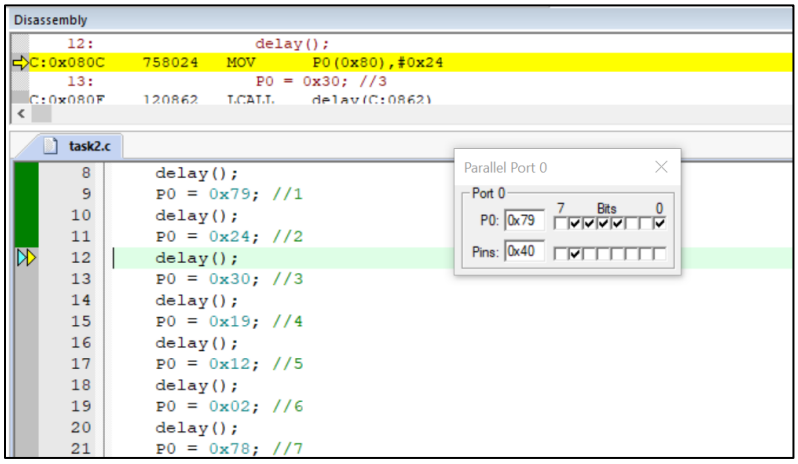
Display digits from 0 to F on Seven Segment Display

Code:

```
#include<reg51.h>
void delay();

void main(void){

  while(1){
    P0 = 0x40; //0
    delay();
    P0 = 0x79; //1
    delay();
    P0 = 0x24; //2
    delay();
    P0 = 0x30; //3
    delay();
    P0 = 0x19; //4
    delay();
    P0 = 0x12; //5
    delay();
    P0 = 0x02; //6
    delay();
    P0 = 0x78; //7
    delay();
    P0 = 0x00; //8
    delay();
```

```
24      delay();
25      P0 = 0x18; //9
26      delay();
27      P0 = 0x08; //A
28      delay();
29      P0 = 0x00; //B
30      delay();
31      P0 = 0x46; //C
32      delay();
33      P0 = 0x40; //D
34      delay();
35      P0 = 0x06; //E
36      delay();
37      P0 = 0x0E; //F
38      delay();
39
40    }
41  }
42  void delay(){
43    unsigned int y, x;
44    for(x = 0; x<30000; x++);
45    for(y = 0; y<30000; y++);
46  }
47
```

Output:



Proteus Simulation: